Kalman Filter Theory and Applications
State Space Concepts

https://github.com/musicarroll/kalman_course

Michael L. Carroll

July 30, 2024

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Part III

State Space Concepts

Topics

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State Space Concepts

Topics

- Understanding the Equations: Heuristic Introduction

Part III

State Space Concepts

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- Understanding the Equations: Heuristic Introduction
- Equation Drilldown: Taking the Equations Apart

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- **State Space Concepts** ($\geq$8 videos)

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State Space Concepts

Topics

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- Dynamic Systems: The Concept of State Space

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- Deriving State Models from Differential Equations

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Simple Harmonic Oscillator

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$$x_1 = y$$
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and $\dot{x}_2 = \ddot{x}_1 = -\frac{k}{m}x_1 - 0 \cdot x_2$.

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- In matrix and vector form:

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- A schematic drawing of this system is depicted on the next slide

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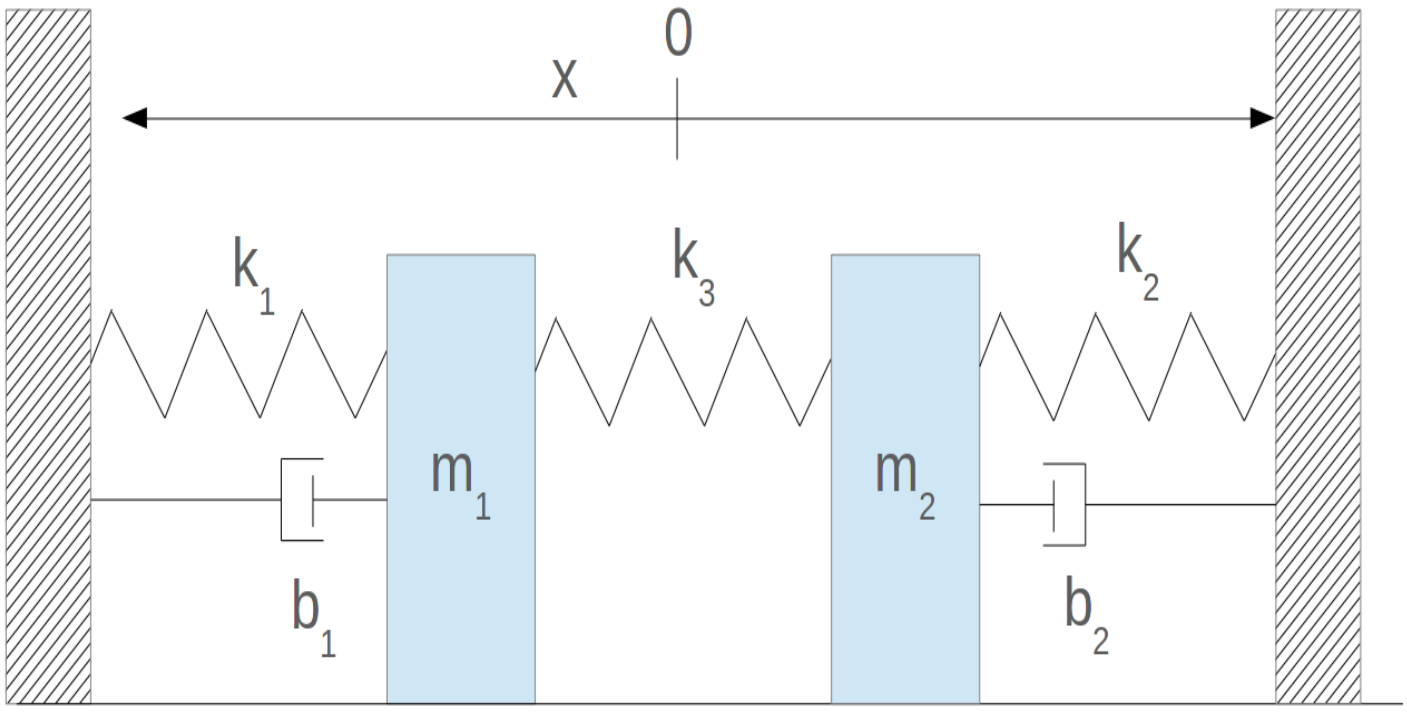
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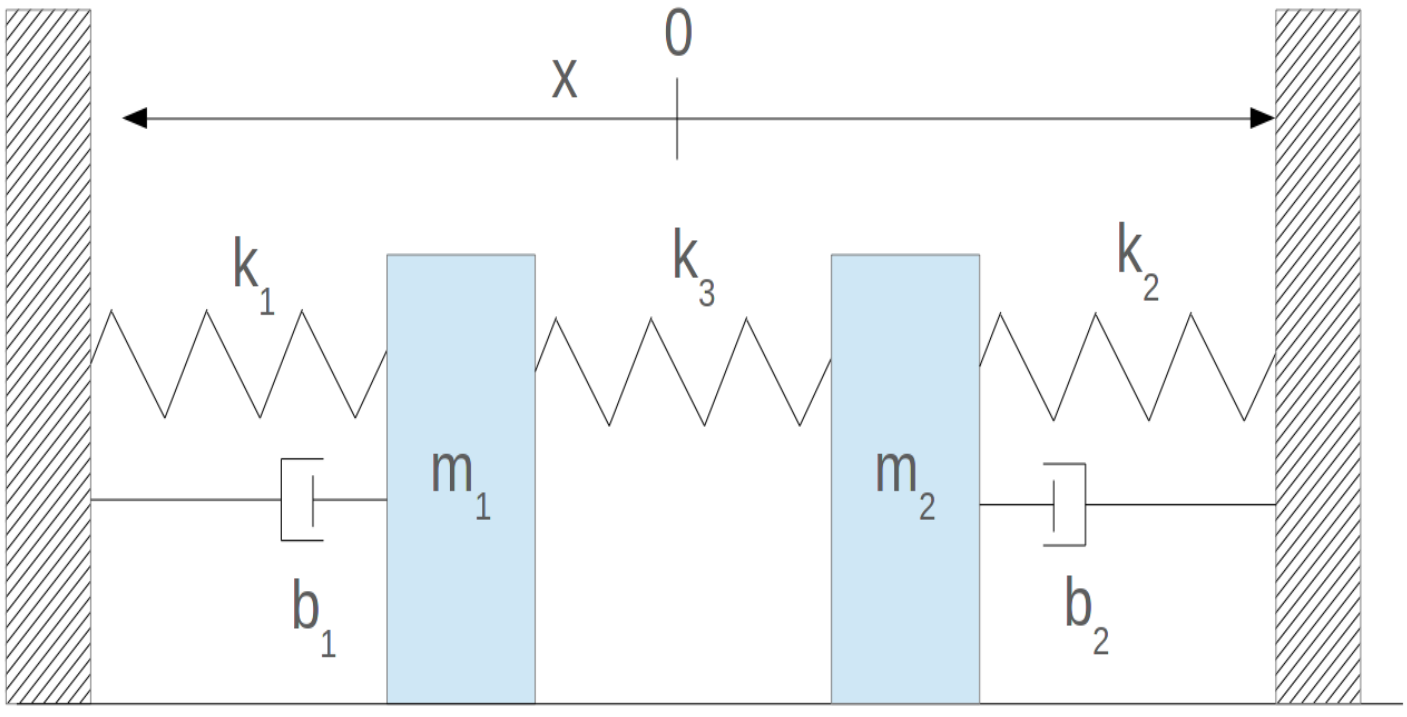
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- The state space equations can be written as:

$$\frac{dx_1}{dt} = x_3$$

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$$\frac{dx_3}{dt} = \frac{-k_1 \cdot x_1 - k_3 \cdot (x_1 - x_2) - b_1 \cdot x_3}{m_1}$$

$$\frac{dx_4}{dt} = \frac{-k_2 \cdot x_2 + k_3 \cdot (x_1 - x_2) - b_2 \cdot x_4}{m_2}$$

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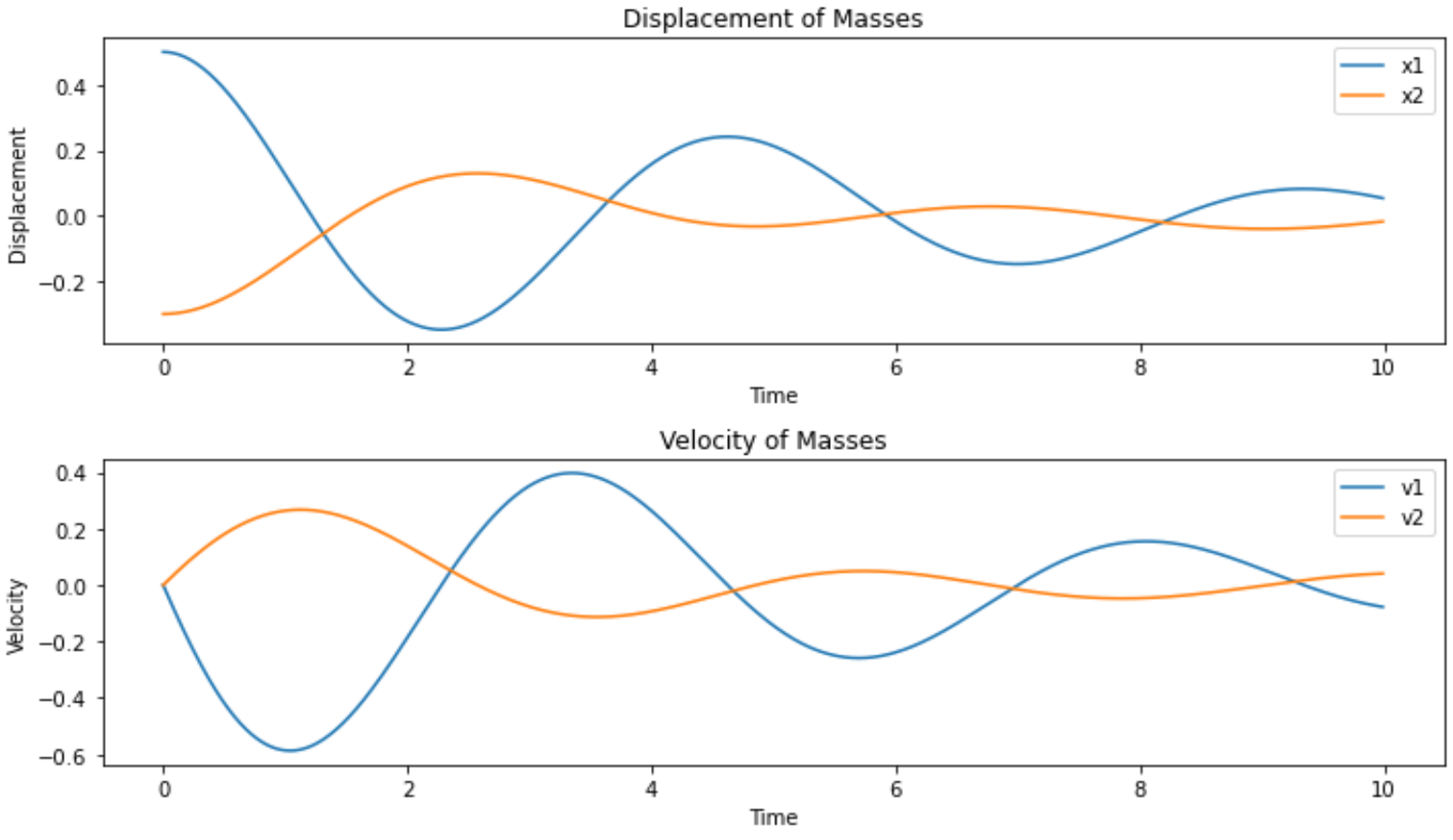
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- A plot of a python simulation of this system is shown in the next slide

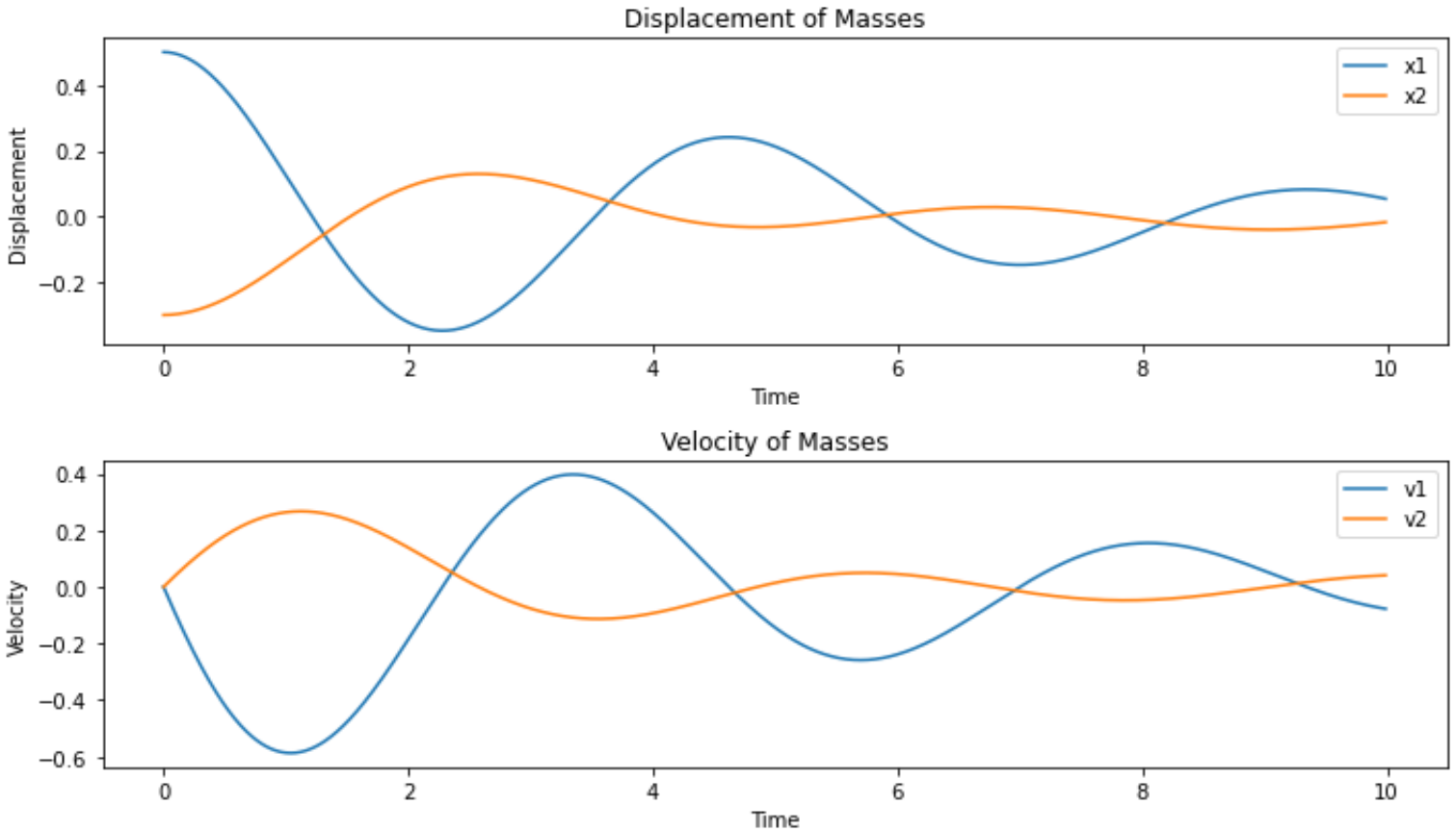
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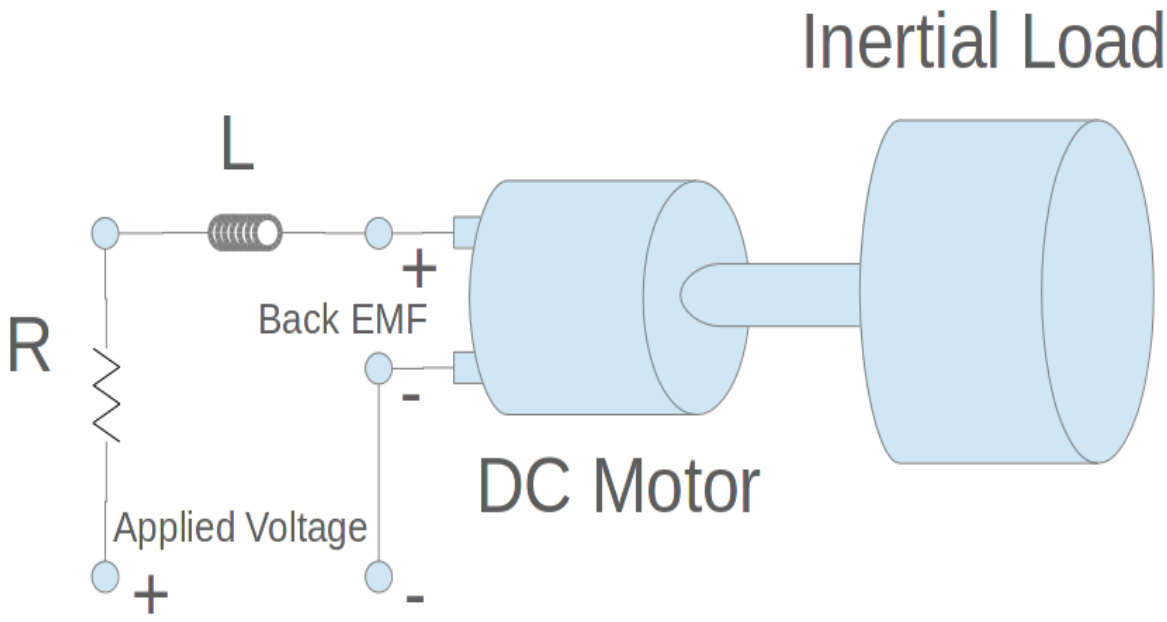
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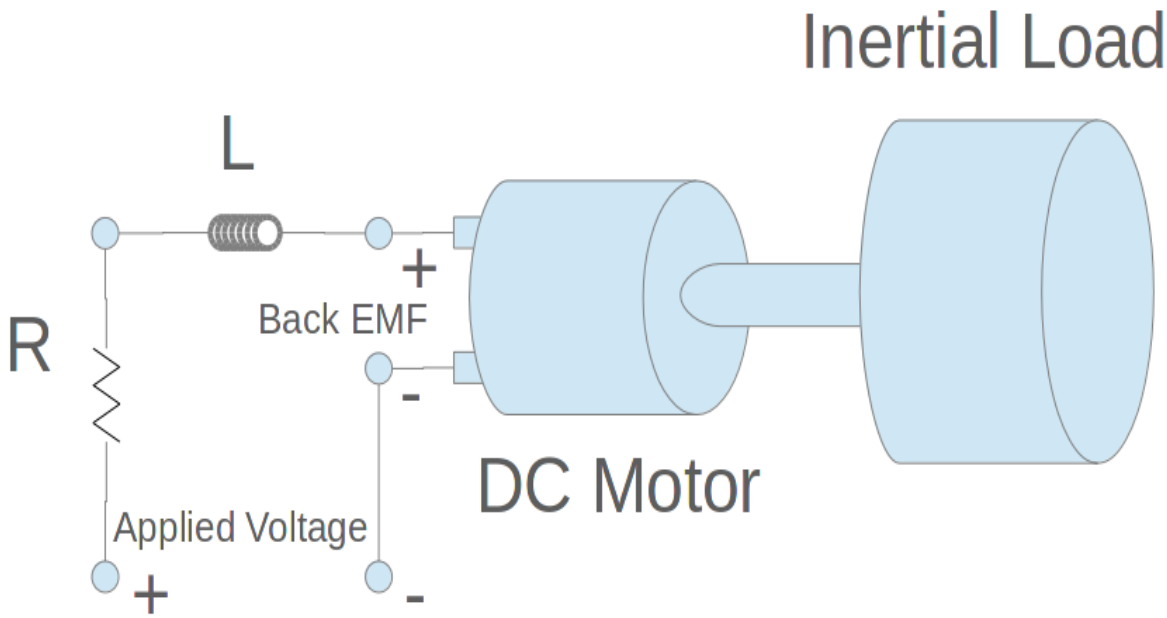
State Space Representation

Example: DC Motor with Inertial Load: An Electromechanical System



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State Space Model

DC Motor with Inertial Load

System Description

Consider a DC motor with an inertial load. The key variables and constants are as follows:

R : Motor resistance

L : Motor inductance

θ : Rotational position of the load

ω : Rate of change of θ

K_f : Frictional constant

τ : Torque

K_m : Armature constant

v_{app} : Applied voltage

v_{emf} : Back EMF

K_b : EMF constant

How the Motor Generates Back EMF

Electromagnetic Induction and Rotational Motion

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State Space Model

DC Motor with Inertial Load

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DC Motor with Inertial Load

State space equations:

$$\frac{d\theta}{dt} = \omega$$

$$L\frac{di}{dt} = v_{\text{app}} - v_{\text{emf}} - Ri$$

$$J\frac{d\omega}{dt} = \tau - K_f\omega$$

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Where:

θ : Rotational position

ω : Angular velocity

i : Current through the motor

J : Moment of inertia of the load

τ : Applied torque

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In matrix form, the state space model can be expressed as:

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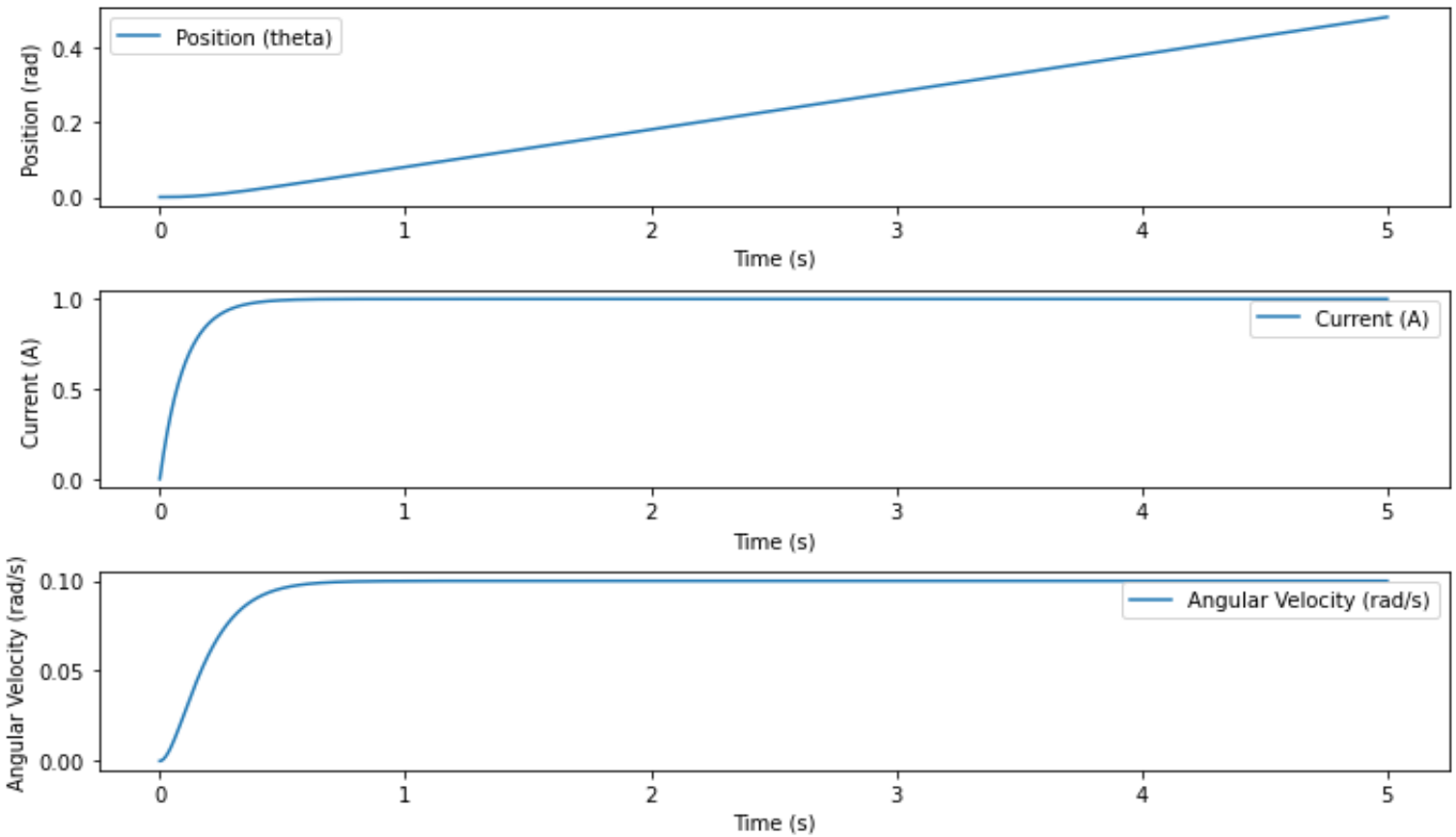
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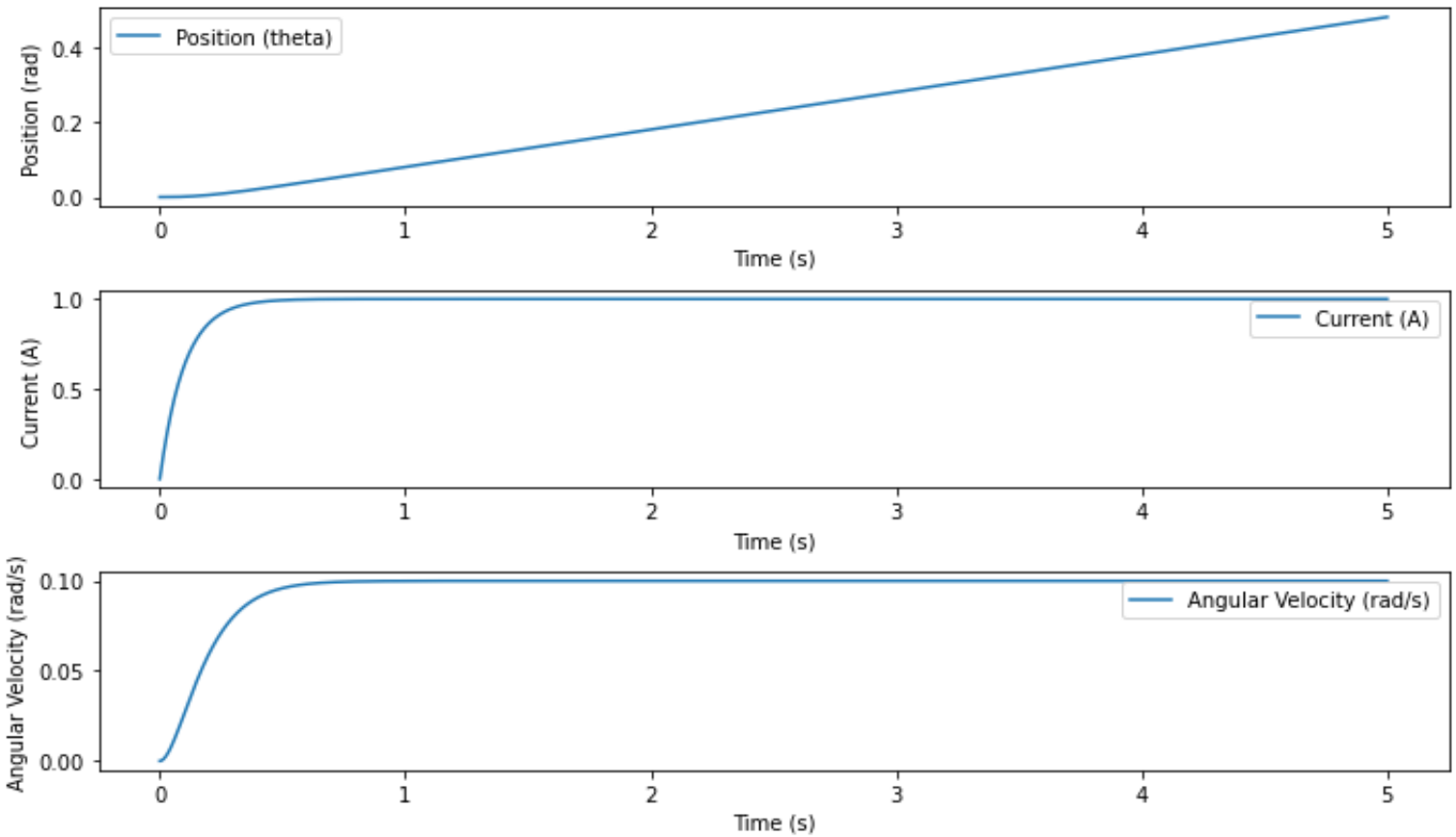
State Space Representation

Example: DC Motor with Inertial Load



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State Space Representation

Predator-Prey Model

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- Underlying Assumptions:

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Coupled ODEs

Predator-Prey Differential Equations

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Coupled differential equation model for Lotka-Volterra predator-prey system:

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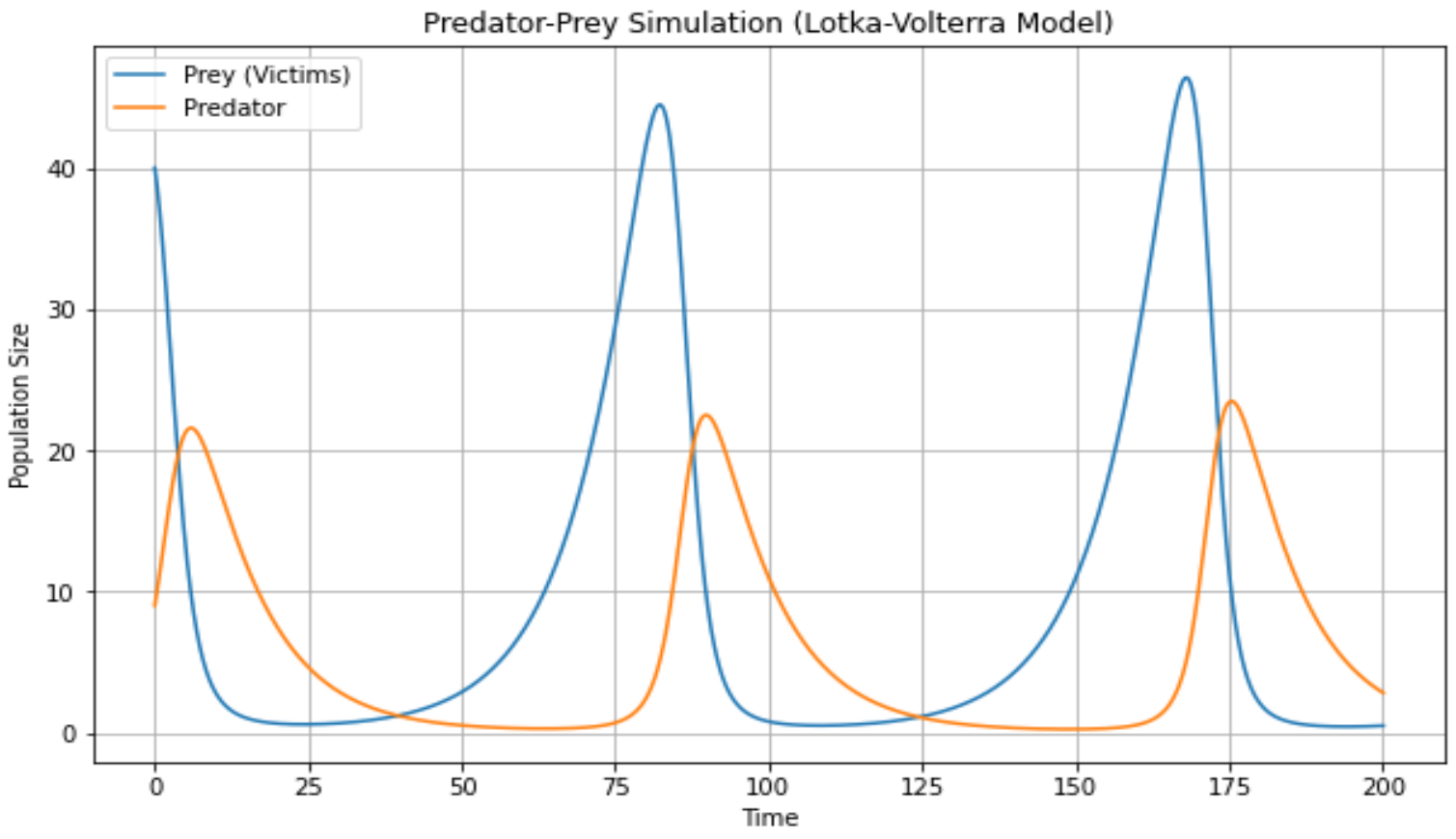
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Example: Predator-Prey Equations, Nominal Initial Conditions

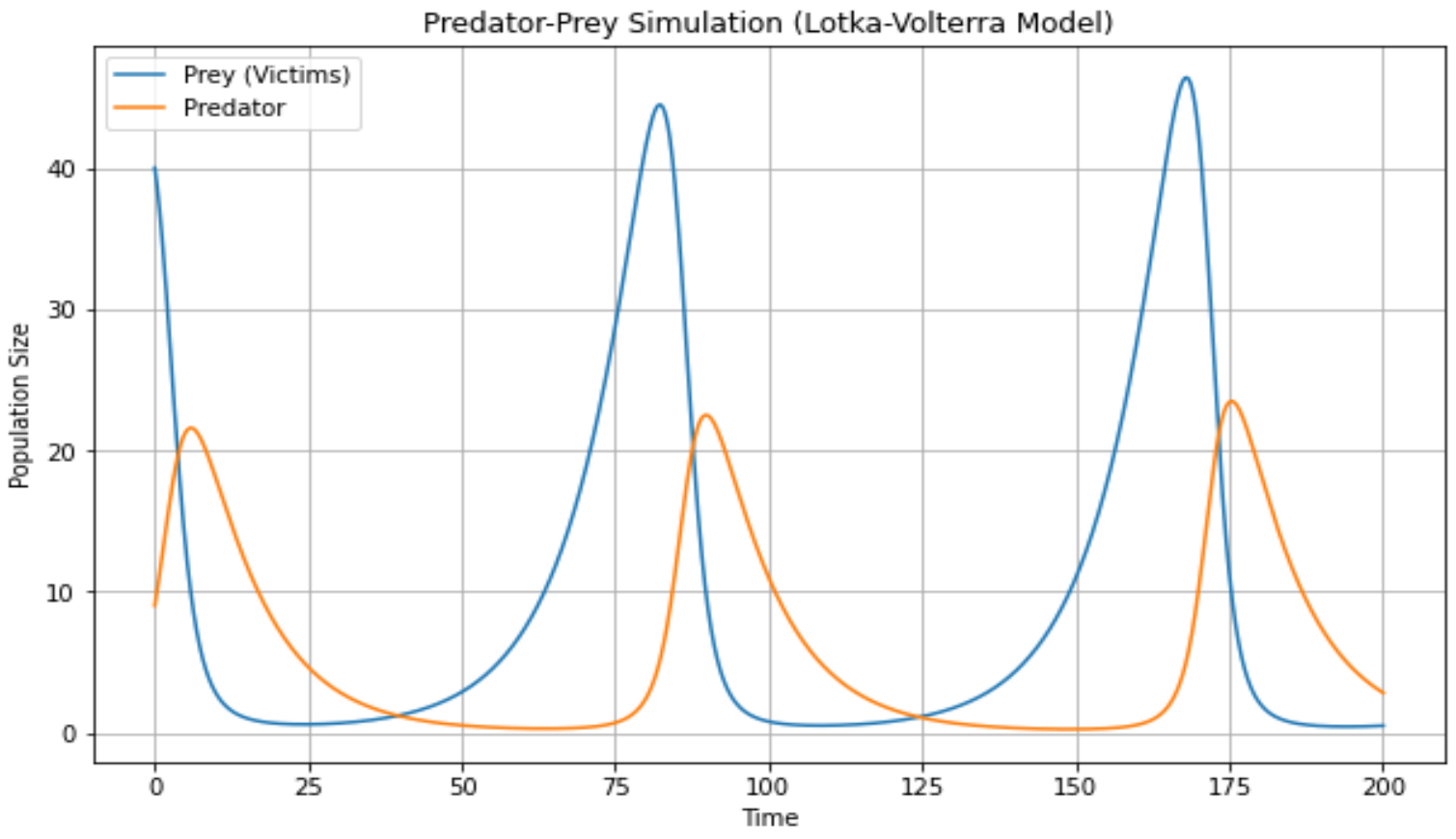
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